

# pycc test-suite timings

psi4 1.9, conv/pk, macOS single machine. Duration = call+setup+teardown.

222 tests with measurable time, total 1424s ( $\approx 24$  min). Local-cc slow tests excluded;  $\sim 12$  further ran  $< 0.005$  s. FD = executes finite difference.

| Test (file::function)                                                             | Mol.    | Basis          | s      | FD |
|-----------------------------------------------------------------------------------|---------|----------------|--------|----|
| test_059_cc3_response::test_cc3_polarizability                                    | H2O     | STO-3G         | 168.15 |    |
| test_037_rtcc3::test_rtcc_he_cc_pvdz                                              | H2O     | cc-pVDZ        | 125.81 |    |
| test_068_mp2_apt::test_so_mp2_corr_apt_vs_spatial_ccpvdz                          | H2O     | cc-pVDZ        | 109.89 |    |
| test_065_spinorbital_hf_hessian::test_so_rhf_hessian_vs_spatial_ccpvdz            | H2O     | cc-pVDZ        | 64.90  |    |
| test_031_cc3::test_so_cc3_lambda_equals_spatial_rhf                               | H2O     | STO-3G         | 63.41  |    |
| test_069_mp2_hessian::test_mp2_hessian_2n1_vs_explicit_631g                       | H2O     | 6-31G          | 43.91  |    |
| test_031_cc3::test_cc3_h2o                                                        | H2O     | cc-pVDZ        | 40.83  |    |
| test_067_mp2_polarizability::test_fc_so_mp2_corr_polarizability_vs_spatial_ccpvdz | H2O     | cc-pVDZ        | 40.38  |    |
| test_031_cc3::test_so_cc3_lambda_equals_spatial_rhf_frzc                          | H2O     | STO-3G         | 31.82  |    |
| test_059_cc3_response::test_cc3_optical_rotation_zz                               | H2O2    | STO-3G         | 31.63  |    |
| test_067_mp2_polarizability::test_so_mp2_corr_polarizability_vs_spatial_ccpvdz    | H2O     | cc-pVDZ        | 30.01  |    |
| test_070_mp2_2n1_polarizability::test_2n1_polarizability_vs_explicit_ccpvdz       | H2O     | cc-pVDZ        | 29.83  |    |
| test_023_ms_int::test_rtcc_water_cc_pvdz                                          | H2O     | cc-pVDZ        | 26.03  |    |
| test_031_cc3::test_so_cc3_equals_spatial_rhf                                      | H2O     | STO-3G         | 22.80  |    |
| test_031_cc3::test_ucc3_oh                                                        | OH      | 6-31G          | 22.11  |    |
| test_054_rohf::test_rohf_cc3_oh                                                   | OH      | 6-31G          | 21.84  |    |
| test_069_mp2_hessian::test_fc_so_mp2_corr_hessian_vs_spatial_631g                 | H2O     | 6-31G          | 20.97  |    |
| test_069_mp2_hessian::test_so_mp2_corr_hessian_vs_spatial_631g                    | H2O     | 6-31G          | 18.88  |    |
| test_068_mp2_apt::test_mp2_apt_2n1_routes_vs_explicit_631g                        | H2O     | 6-31G          | 17.31  |    |
| test_071_hf_velocity_apt::test_hf_vgapt_so_vs_spatial                             | NH3     | 6-31G*/6-31G** | 16.17  |    |
| test_074_property_facade::test_facade_route_option                                | H2O2    | STO-3G         | 15.28  |    |
| test_034_ccsd_t_density::test_ccsd_t_h2o                                          | H2O     | STO-3G/cc-pVDZ | 14.43  |    |
| test_061_mp2_relaxed_density::test_mp2_explicit_corr_gradient_ccpvdz              | H2O     | cc-pVDZ        | 13.13  |    |
| test_031_cc3::test_so_cc3_equals_spatial_rhf_frzc                                 | H2O     | STO-3G         | 11.98  |    |
| test_072_mp2_aat::test_mp2_aat_so_equals_spatial                                  | H2O2    | STO-3G         | 11.60  |    |
| test_074_property_facade::test_facade_decomposition_and_total                     | H2O2    | STO-3G         | 11.10  |    |
| test_068_mp2_apt::test_fc_so_mp2_corr_apt_vs_spatial_631g                         | H2O     | 6-31G          | 9.98   |    |
| test_005_ccsd_t_energy::test_ccsd_t_h2o                                           | H2O     | STO-3G/cc-pVDZ | 9.66   |    |
| test_022_adap_int::test_rtcc_water_cc_pvdz                                        | H2O     | cc-pVDZ        | 9.34   |    |
| test_043_eomccsd::test_eomccsd_c2h4_fc                                            | C2H4    | cc-pVDZ        | 8.89   |    |
| test_061_mp2_relaxed_density::test_fc_mp2_gradient_vs_energy_fd_631g              | H2O     | 6-31G          | 8.81   | •  |
| test_036_lr::test_linresp                                                         | H2O     | aug-cc-pvdz    | 8.49   |    |
| test_064_spinorbital_hf_apt::test_so_rhf_apt_vs_spatial_ccpvdz                    | H2O     | cc-pVDZ        | 7.64   |    |
| test_068_mp2_apt::test_so_mp2_corr_apt_vs_spatial_631g                            | H2O     | 6-31G          | 7.58   |    |
| test_058_spatial_optrot::test_spatial_rhf_optrot_vs_psi4                          | H2O2    | cc-pVDZ        | 7.58   |    |
| test_065_spinorbital_hf_hessian::test_so_rhf_hessian_vs_spatial_631g              | H2O     | 6-31G          | 7.45   |    |
| test_068_mp2_apt::test_mp2_corr_apt_dipfd_631g                                    | H2O     | 6-31G          | 6.90   | •  |
| test_069_mp2_hessian::test_fc_mp2_corr_hessian_gradfd_631g                        | H2O     | 6-31G          | 6.72   | •  |
| test_069_mp2_hessian::test_mp2_corr_hessian_gradfd_631g                           | H2O     | 6-31G          | 6.48   | •  |
| test_019_localrt::test_rtpao                                                      | H2O     | cc-pVDZ        | 6.18   |    |
| test_019_localrt::test_rtpno                                                      | H2O     | cc-pVDZ        | 6.12   |    |
| test_061_mp2_relaxed_density::test_mp2_explicit_corr_dipole_ccpvdz                | H2O     | cc-pVDZ        | 6.01   | •  |
| test_067_mp2_polarizability::test_fc_mp2_corr_polarizability_dipfd_631g           | H2O     | 6-31G          | 5.90   | •  |
| test_068_mp2_apt::test_fc_mp2_corr_apt_dipfd_631g                                 | H2O     | 6-31G          | 5.89   | •  |
| test_075_spinorbital_eomccsd::test_so_eom_closed_shell_keystone                   | H2O     | STO-3G         | 5.69   |    |
| test_054_rohf::test_rohf_mp2_ccsd_ccsd_t_oh                                       | OH      | cc-pVDZ        | 5.39   |    |
| test_061_mp2_relaxed_density::test_mp2_gradient_ccpvdz                            | H2O     | cc-pVDZ        | 5.36   |    |
| test_067_mp2_polarizability::test_mp2_corr_polarizability_dipfd_631g              | H2O     | 6-31G          | 5.24   | •  |
| test_073_mp2_vg_apt::test_mp2_vg_apt_gauge_invariance                             | H2O2    | STO-3G         | 5.11   |    |
| test_073_mp2_vg_apt::test_mp2_vg_apt_so_equals_spatial                            | H2O2    | STO-3G         | 4.82   |    |
| test_072_mp2_aat::test_mp2_aat_gauge_invariance                                   | H2O2    | STO-3G         | 4.77   |    |
| test_067_mp2_polarizability::test_ump2_corr_polarizability_energy_fd_nh2_631g     | H2O/NH2 | 6-31G          | 4.51   | •  |
| test_074_property_facade::test_facade_hf_wavefunction                             | H2O2    | STO-3G         | 4.47   |    |
| test_005_ccsd_t_energy::test_uccsd_t_oh                                           | OH      | cc-pVDZ        | 4.41   |    |
| test_007_dipole::test_so_dipole_equals_spatial_rhf                                | H2O     | cc-pVDZ        | 4.23   |    |
| test_026_autocorrelation::test_rtcc_he_cc_pvdz_auto                               | He      | cc-pVDZ        | 4.15   |    |
| test_057_spatial_polar::test_spatial_rhf_polar_vs_psi4                            | H2O     | cc-pVDZ        | 3.89   |    |
| test_006_rtccsd::test_rtcc_he_cc_pvdz                                             | He      | cc-pVDZ        | 3.82   |    |
| test_067_mp2_polarizability::test_fc_so_mp2_corr_polarizability_vs_spatial_631g   | H2O     | 6-31G          | 3.56   |    |
| test_048_dipole_derivatives::test_dipole_derivatives_h2o                          | H2O     | STO-3G         | 3.51   | •  |

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| Test (file::function)                                                        | Mol.    | Basis          | s    | FD |
|------------------------------------------------------------------------------|---------|----------------|------|----|
| test_069_mp2_hessian::test_fc_mp2_corr_hessian_symmetry_sum_rule_631g        | H2O     | 6-31G          | 3.18 |    |
| test_005_ccsd_t_energy::test_uccsd_t_oh_frzc                                 | OH      | cc-pVDZ        | 3.08 |    |
| test_058_spatial_optrot::test_spatial_rhf_optrot_vs_spinorbital              | H2O2    | STO-3G         | 3.06 |    |
| test_064_spinorbital_hf_apt::test_uhf_apt_vs_finite_difference               | OH      | 6-31G          | 3.05 | •  |
| test_021_rk4::test_rtcc_water_cc_pvdz                                        | H2O     | cc-pVDZ        | 3.01 |    |
| test_056_spinorbital_optrot::test_so_rhf_optrot_vs_psi4                      | H2O2    | STO-3G         | 2.96 |    |
| test_062_spinorbital_hf_gradient::test_so_rhf_gradient_vs_spatial_ccpvdz     | H2O     | cc-pVDZ        | 2.85 |    |
| test_061_mp2_relaxed_density::test_mp2_relaxed_dipole_ccpvdz                 | H2O     | cc-pVDZ        | 2.82 | •  |
| test_071_hf_velocity_apt::test_hf_vgapt_vs_amos                              | NH3     | 6-31G*/6-31G** | 2.69 |    |
| test_065_spinorbital_hf_hessian::test_uhf_hessian_vs_psi4                    | OH      | 6-31G          | 2.64 |    |
| test_067_mp2_polarizability::test_mp2_corr_polarizability_energy_fd_631g     | H2O     | 6-31G          | 2.60 | •  |
| test_002_ccsd_energy::test_uccsd_oh                                          | OH      | cc-pVDZ        | 2.55 |    |
| test_069_mp2_hessian::test_mp2_corr_hessian_symmetry_sum_rule_631g           | H2O     | 6-31G          | 2.37 |    |
| test_070_mp2_2n1_polarizability::test_2n1_polarizability_vs_explicit_631g    | H2O     | 6-31G          | 2.36 |    |
| test_002_ccsd_energy::test_uccsd_oh_frzc                                     | OH      | cc-pVDZ        | 2.34 |    |
| test_059_cc3_response::test_cc3_polarizability_zz_frzc                       | H2O     | STO-3G         | 2.34 | •  |
| test_024_contract_cpu::test_rtcc_water_cc_pvdz                               | H2O     | cc-pVDZ        | 2.33 |    |
| test_068_mp2_apt::test_mp2_nuclear_t2_response_631g                          | H2O     | 6-31G          | 2.29 | •  |
| test_061_mp2_relaxed_density::test_ump2_relaxed_dipole_nh2_631g              | NH2     | 6-31G          | 2.29 | •  |
| test_070_mp2_2n1_polarizability::test_2n1_polarizability_vs_explicit_fc_631g | H2O     | 6-31G          | 2.26 |    |
| test_078_so_ccsd_gradient::test_so_ccsd_gradient_vs_psi4_uhf_oh_c2v          | OH      | 6-31G          | 2.24 |    |
| test_027_linresp::test_polar_h2o_cc_pvdz                                     | H2O     | STO-3G         | 2.15 |    |
| test_066_spinorbital_hf_aat::test_so_rhf_aat_vs_spatial_h2o2                 | H2O2    | sto-3g         | 2.10 |    |
| test_061_mp2_relaxed_density::test_mp2_explicit_corr_dipole_631g             | H2O     | 6-31G          | 2.10 | •  |
| test_061_mp2_relaxed_density::test_mp2_relaxed_dipole_631g                   | H2O     | 6-31G          | 2.04 | •  |
| test_061_mp2_relaxed_density::test_fc_so_mp2_explicit_corr_dipole_631g       | H2O     | 6-31G          | 2.04 | •  |
| test_074_property_facade::test_aat_frozen_core_total                         | H2O2    | STO-3G         | 2.00 |    |
| test_067_mp2_polarizability::test_so_mp2_corr_polarizability_vs_spatial_631g | H2O     | 6-31G          | 1.98 |    |
| test_061_mp2_relaxed_density::test_fc_mp2_relaxed_dipole_631g                | H2O     | 6-31G          | 1.96 | •  |
| test_035_cpnoppcc::test_pnopp_ccsd_opt                                       | H2O     | cc-pVDZ        | 1.96 |    |
| test_061_mp2_relaxed_density::test_fc_so_mp2_relaxed_dipole_631g             | H2O     | 6-31G          | 1.95 | •  |
| test_061_mp2_relaxed_density::test_fc_sa_mp2_explicit_corr_dipole_631g       | H2O     | 6-31G          | 1.92 | •  |
| test_061_mp2_relaxed_density::test_sa_mp2_relaxed_dipole_631g                | H2O     | 6-31G          | 1.91 | •  |
| test_003_ccsd_lambda::test_uhf_lambda_vs_cfour                               | OH      | cc-pVDZ        | 1.90 |    |
| test_016_chk::test_chk                                                       | H2      | cc-pVDZ        | 1.90 |    |
| test_061_mp2_relaxed_density::test_sa_mp2_explicit_corr_dipole_631g          | H2O     | 6-31G          | 1.89 | •  |
| test_007_dipole::test_uhf_dipole_vs_cfour                                    | OH      | cc-pVDZ        | 1.88 |    |
| test_003_ccsd_lambda::test_rohf_lambda_vs_cfour                              | OH      | cc-pVDZ        | 1.88 |    |
| test_007_dipole::test_rohf_dipole_vs_cfour                                   | OH      | cc-pVDZ        | 1.82 |    |
| test_059_cc3_response::test_cc3_polarizability_zz                            | H2O     | STO-3G         | 1.81 |    |
| test_074_property_facade::test_facade_orbital_gauge_option                   | H2O2    | STO-3G         | 1.79 |    |
| test_005_ccsd_t_energy::test_so_ccsd_t_equals_spatial_rhf                    | H2O     | STO-3G         | 1.79 |    |
| test_060_field_energies::test_cc3_field_findiff                              | H2O     | STO-3G         | 1.78 | •  |
| test_070_mp2_2n1_polarizability::test_2n1_polarizability_uhf_nh2_631g        | H2O/NH2 | 6-31G          | 1.77 |    |
| test_071_hf_velocity_apt::test_hf_vgapt_origin_invariant                     | NH3     | 6-31G*         | 1.76 |    |
| test_035_cpnoppcc::test_cpnopp_ccsd                                          | H2O     | cc-pVDZ        | 1.73 |    |
| test_075_spinorbital_eomccsd::test_so_eom_sigma_vs_psi4_uhf                  | OH      | STO-3G         | 1.70 |    |
| test_055_spinorbital_polar::test_so_rhf_polar_vs_psi4                        | H2O     | STO-3G         | 1.68 |    |
| test_061_mp2_relaxed_density::test_sa_mp2_gradient_ccpvdz                    | H2O     | cc-pVDZ        | 1.64 |    |
| test_068_mp2_apt::test_fc_mp2_apt_translational_sum_rule_631g                | H2O     | 6-31G          | 1.61 |    |
| test_043_eomccsd::test_eomccsd_h2o                                           | H2O     | cc-pVDZ        | 1.52 |    |
| test_074_property_facade::test_aat_genuine_separation                        | H2O2    | STO-3G         | 1.50 |    |
| test_061_mp2_relaxed_density::test_fc_so_mp2_explicit_corr_gradient_631g     | H2O     | 6-31G          | 1.47 |    |
| test_061_mp2_relaxed_density::test_mp2_explicit_corr_gradient_631g           | H2O     | 6-31G          | 1.41 |    |
| test_028_pnoppcc::test_pnopp_ccsd_opt                                        | H2O     | cc-pVDZ        | 1.39 |    |
| test_051_cisd::test_rohf_cisd_cid_vs_cfour                                   | OH      | cc-pVDZ        | 1.34 |    |
| test_061_mp2_relaxed_density::test_fc_mp2_gradient_spatial_vs_so_631g        | H2O     | 6-31G          | 1.31 |    |
| test_043_eomccsd::test_eomccsd_h2o_fc                                        | H2O     | cc-pVDZ        | 1.31 |    |
| test_073_mp2_vg_apt::test_mp2_vg_apt_reduces_to_hf                           | H2O2    | STO-3G         | 1.30 |    |
| test_078_so_ccsd_gradient::test_so_ccsd_gradient_vs_psi4_uhf                 | NH2     | STO-3G         | 1.27 |    |
| test_056_spinorbital_optrot::test_rohf_optrot_runs                           | OH      | STO-3G         | 1.26 |    |
| test_068_mp2_apt::test_mp2_apt_translational_sum_rule_631g                   | H2O     | 6-31G          | 1.24 |    |
| test_051_cisd::test_uhf_cisd_cid_vs_cfour                                    | OH      | cc-pVDZ        | 1.24 |    |
| test_061_mp2_relaxed_density::test_mp2_gradient_spatial_vs_so_631g           | H2O     | 6-31G          | 1.23 |    |
| test_056_spinorbital_optrot::test_uhf_optrot_runs                            | OH      | STO-3G         | 1.23 |    |
| test_061_mp2_relaxed_density::test_mp2_total_dipole_631g                     | H2O     | 6-31G          | 1.22 | •  |
| test_061_mp2_relaxed_density::test_ump2_gradient_nh2_631g                    | NH2     | 6-31G          | 1.19 |    |
| test_055_spinorbital_polar::test_uhf_polar_findiff                           | OH      | STO-3G         | 1.19 | •  |

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| Test (file::function)                                                         | Mol.   | Basis          | s    | FD |
|-------------------------------------------------------------------------------|--------|----------------|------|----|
| test_064_spinorbital_hf_apt::test_so_rhf_apt_vs_spatial_631g                  | H2O    | 6-31G          | 1.18 |    |
| test_066_spinorbital_hf_aat::test_so_rhf_aat_vs_spatial_631g                  | H2O    | 6-31G          | 1.15 |    |
| test_070_mp2_2n1_polarizability::test_2n1_polarizability_so_vs_spatial_631g   | H2O    | 6-31G          | 1.14 |    |
| test_074_property_facade::test_aat_decomposition_identities                   | H2O2   | STO-3G         | 1.13 |    |
| test_072_mp2_aat::test_mp2_aat_frozen_core                                    | H2O2   | STO-3G         | 1.12 |    |
| test_073_mp2_vg_apt::test_mp2_vg_apt_frozen_core                              | H2O2   | STO-3G         | 1.11 |    |
| test_057_spatial_polar::test_spatial_rhf_polar_vs_spinorbital                 | H2O    | STO-3G         | 1.10 |    |
| test_055_spinorbital_polar::test_rohf_polar_findiff                           | OH     | STO-3G         | 1.09 | •  |
| test_061_mp2_relaxed_density::test_mp2_gradient_631g                          | H2O    | 6-31G          | 1.08 |    |
| test_061_mp2_relaxed_density::test_mp2_explicit_equals_relaxed_631g           | H2O    | 6-31G          | 1.05 |    |
| test_046_hf_gradient::test_hf_gradient_h2o                                    | H2O    | cc-pVDZ        | 1.03 |    |
| test_013_pnocc::test_pno_ccsd_opt                                             | H2O    | cc-pVDZ        | 1.01 |    |
| test_072_mp2_aat::test_mp2_aat_all_electron                                   | H2O2   | STO-3G         | 1.01 |    |
| test_074_property_facade::test_aat_origin_argument                            | H2O2   | STO-3G         | 1.01 |    |
| test_028_pnopcc::test_pnopccsd                                                | H2O    | cc-pVDZ        | 0.99 |    |
| test_046_hf_gradient::test_hf_gradient_frozen_core_ref                        | H2O    | cc-pVDZ        | 0.99 |    |
| test_074_property_facade::test_aat_nuclear_term                               | H2O2   | STO-3G         | 0.98 |    |
| test_075_spinorbital_eomccsd::test_so_eom_left_equals_right                   | H2O/OH | STO-3G         | 0.93 |    |
| test_076_ccsd_gradient::test_ccsd_gradient_frozen_core                        | H2O    | STO-3G         | 0.92 |    |
| test_076_ccsd_gradient::test_ccsd_gradient_vs_psi4                            | H2O    | STO-3G         | 0.92 |    |
| test_073_mp2_vg_apt::test_mp2_vg_apt_all_electron                             | H2O2   | STO-3G         | 0.91 |    |
| test_013_pnocc::test_pno_ccsd                                                 | H2O    | cc-pVDZ        | 0.91 |    |
| test_052_spinorbital_hamiltonian::test_so_equals_spatial_rhf_frozen_core      | H2O    | cc-pVDZ        | 0.91 |    |
| test_052_spinorbital_hamiltonian::test_so_equals_spatial_rhf                  | H2O    | cc-pVDZ        | 0.90 |    |
| test_063_spinorbital_hf_polarizability::test_so_rhf_polarizability_vs_spatial | H2O    | cc-pVDZ        | 0.89 |    |
| test_004_ccsd_density::test_density_ccsd_h2o                                  | H2O    | STO-3G/cc-pVDZ | 0.87 |    |
| test_071_hf_velocity_apt::test_hf_vgapt_differs_from_lg_finite_basis          | NH3    | 6-31G*         | 0.86 |    |
| test_060_field_energies::test_ccsd_field_findiff                              | H2O    | STO-3G         | 0.85 | •  |
| test_030_sp::test_rtcc_water_cc_pvdz                                          | H2O    | cc-pvdz        | 0.85 |    |
| test_003_ccsd_lambda::test_lambda_ccsd_h2o                                    | H2O    | STO-3G/cc-pVDZ | 0.82 |    |
| test_049_hf_hessian::test_hf_hessian_h2o                                      | H2O    | STO-3G         | 0.79 |    |
| test_062_spinorbital_hf_gradient::test_so_rhf_gradient_vs_spatial_631g        | H2O    | 6-31G          | 0.79 |    |
| test_002_ccsd_energy::test_ccsd_h2o                                           | H2O    | STO-3G/cc-pVDZ | 0.78 |    |
| test_052_spinorbital_hamiltonian::test_uhf_so_mp2                             | OH     | cc-pVDZ        | 0.75 |    |
| test_078_so_ccsd_gradient::test_so_ccsd_gradient_frozen_core                  | H2O    | STO-3G         | 0.74 |    |
| test_077_so_ccsd_2pdm::test_so_2pdm_reconstructs_energy_frozen_core           | H2O    | STO-3G         | 0.68 |    |
| test_078_so_ccsd_gradient::test_so_ccsd_gradient_zvector_equals_explicit      | H2O    | STO-3G         | 0.67 |    |
| test_077_so_ccsd_2pdm::test_so_2pdm_reconstructs_energy_uhf                   | NH2/OH | STO-3G         | 0.66 |    |
| test_045_mp2::test_mp2_h2o_frzc                                               | H2O    | cc-pVDZ        | 0.64 |    |
| test_045_mp2::test_ump2_oh                                                    | OH     | cc-pVDZ        | 0.64 |    |
| test_045_mp2::test_mp2_h2o                                                    | H2O    | cc-pVDZ        | 0.64 |    |
| test_061_mp2_relaxed_density::test_hf_dipole_ccpvdz                           | H2O    | cc-pVDZ        | 0.62 |    |
| test_045_mp2::test_ump2_oh_frzc                                               | OH     | cc-pVDZ        | 0.62 |    |
| test_061_mp2_relaxed_density::test_sa_mp2_gradient_631g                       | H2O    | 6-31G          | 0.61 |    |
| test_051_cisd::test_so_cisd_cid_equals_spatial_rhf                            | H2O    | 6-31G          | 0.61 |    |
| test_047_cphf_polarizability::test_cphf_polarizability_h2o                    | H2O    | cc-pVDZ        | 0.60 |    |
| test_020_cc2::test_cc2_h2o                                                    | H2O    | cc-pVDZ        | 0.59 |    |
| test_074_property_facade::test_aat_hf_wavefunction                            | H2O2   | STO-3G         | 0.58 |    |
| test_061_mp2_relaxed_density::test_fc_sa_mp2_explicit_corr_gradient_631g      | H2O    | 6-31G          | 0.56 |    |
| test_050_aat::test_aat_h2o2_vs_dalton                                         | H2O2   | STO-3G         | 0.56 |    |
| test_061_mp2_relaxed_density::test_sa_mp2_explicit_corr_gradient_631g         | H2O    | 6-31G          | 0.55 |    |
| test_066_spinorbital_hf_aat::test_uhf_aat_runs                                | OH     | 6-31G          | 0.54 |    |
| test_078_so_ccsd_gradient::test_so_ccsd_gradient_keystone_equals_spatial      | H2O    | STO-3G         | 0.54 |    |
| test_062_spinorbital_hf_gradient::test_uhf_gradient_vs_psi4                   | OH     | 6-31G          | 0.50 |    |
| test_075_spinorbital_eomccsd::test_so_eom_frozen_core                         | OH     | STO-3G         | 0.50 |    |
| test_075_spinorbital_eomccsd::test_so_eom_rohf                                | OH     | STO-3G         | 0.50 |    |
| test_062_spinorbital_hf_gradient::test_rohf_gradient_vs_psi4                  | OH     | 6-31G          | 0.49 |    |
| test_072_mp2_aat::test_mp2_aat_normalization                                  | H2O2   | STO-3G         | 0.49 |    |
| test_065_spinorbital_hf_hessian::test_rohf_hessian_not_implemented            | OH     | 6-31G          | 0.48 |    |
| test_051_cisd::test_cisd_h2o                                                  | H2O    | cc-pVDZ        | 0.47 |    |
| test_051_cisd::test_cisd_h2o_frozen_core                                      | H2O    | cc-pVDZ        | 0.46 |    |
| test_017_ccd::test_ccd_h2o                                                    | H2O    | cc-pVDZ        | 0.46 |    |
| test_063_spinorbital_hf_polarizability::test_uhf_polarizability_vs_psi4       | OH     | 6-31G          | 0.43 |    |
| test_049_hf_hessian::test_hf_hessian_shares_nuclear_cache                     | H2O    | STO-3G         | 0.43 |    |
| test_051_cisd::test_cid_h2o_frozen_core                                       | H2O    | cc-pVDZ        | 0.43 |    |
| test_003_ccsd_lambda::test_so_lambda_equals_spatial_rhf                       | H2O    | STO-3G         | 0.42 |    |
| test_051_cisd::test_cid_h2o                                                   | H2O    | cc-pVDZ        | 0.41 |    |
| test_077_so_ccsd_2pdm::test_so_2pdm_reconstructs_energy_rhf                   | H2O    | STO-3G         | 0.41 |    |

continued...

| Test (file::function)                                                            | Mol.  | Basis   | s    | FD |
|----------------------------------------------------------------------------------|-------|---------|------|----|
| test_007_dipole::test_dipole_h2_2_cc_pvdz                                        | (H2)2 | cc-pVDZ | 0.40 |    |
| test_018_pao::test_pao_H8                                                        | H8    | cc-pVDZ | 0.38 |    |
| test_077_so_ccsd_2pdm::test_so_gradient_densities_convention                     | NH2   | STO-3G  | 0.38 |    |
| test_075_spinorbital_eomccsd::test_so_eom_open_shell_iterative                   | OH    | STO-3G  | 0.38 |    |
| test_076_ccsd_gradient::test_ccsd_gradient_zvector_equals_explicit               | H2O   | STO-3G  | 0.37 |    |
| test_052_spinorbital_hamiltonian::test_auto_dispatch_rhf_is_spatial              | H2O   | cc-pVDZ | 0.37 |    |
| test_064_spinorbital_hf_apt::test_rohf_apt_not_implemented                       | OH    | 6-31G   | 0.35 |    |
| test_002_ccsd_energy::test_so_ccsd_equals_spatial_rhf                            | H2O   | STO-3G  | 0.34 |    |
| test_077_so_ccsd_2pdm::test_so_2pdm_full_assembly                                | OH    | STO-3G  | 0.34 |    |
| test_018_pao::test_pao_h2o                                                       | H2O   | 6-31g   | 0.32 |    |
| test_076_ccsd_gradient::test_ccsd_gradient_correlation_nonzero                   | H2O   | STO-3G  | 0.32 |    |
| test_074_property_facade::test_facade_polarizability_no_nuclear                  | H2O2  | STO-3G  | 0.30 |    |
| test_014_field::test_dipole_h2_2_field                                           | (H2)2 | 6-31G   | 0.30 |    |
| test_076_ccsd_gradient::test_ccsd_gradient_density_energy                        | H2O   | STO-3G  | 0.29 |    |
| test_078_so_ccsd_gradient::test_so_ccsd_gradient_rohf_raises                     | NH2   | STO-3G  | 0.28 |    |
| test_061_mp2_relaxed_density::test_hf_dipole_631g                                | H2O   | 6-31G   | 0.26 |    |
| test_074_property_facade::test_apt_bad_gauge                                     | H2O2  | STO-3G  | 0.25 |    |
| test_070_mp2_2n1_polarizability::test_2n1_polarizability_guards                  | H2O   | 6-31G   | 0.24 |    |
| test_020_cc2::test_cc2_h2                                                        | H2    | cc-pVDZ | 0.21 |    |
| test_066_spinorbital_hf_aat::test_rohf_aat_not_implemented                       | OH    | 6-31G   | 0.20 |    |
| test_063_spinorbital_hf_polarizability::test_rohf_polarizability_not_implemented | OH    | 6-31G   | 0.20 |    |
| test_051_cisd::test_uhf_cisd_equals_fci                                          | H2    | 6-31G   | 0.18 |    |
| test_060_field_energies::test_cc3_field_requires_store_triples                   | H2O   | STO-3G  | 0.17 |    |
| test_051_cisd::test_rohf_cisd_equals_fci                                         | H2    | 6-31G   | 0.17 |    |
| test_009_pade::test_pade_sanity                                                  | —     | —       | 0.01 |    |
| test_009_pade::test_pade_sanity_xfail                                            | —     | —       | 0.01 |    |